

ProgRec: Undergraduate Research Program Mentor Recommend Agent

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The "First-Step" Challenge: Bridging the Undergraduate Research Gap

Problems:

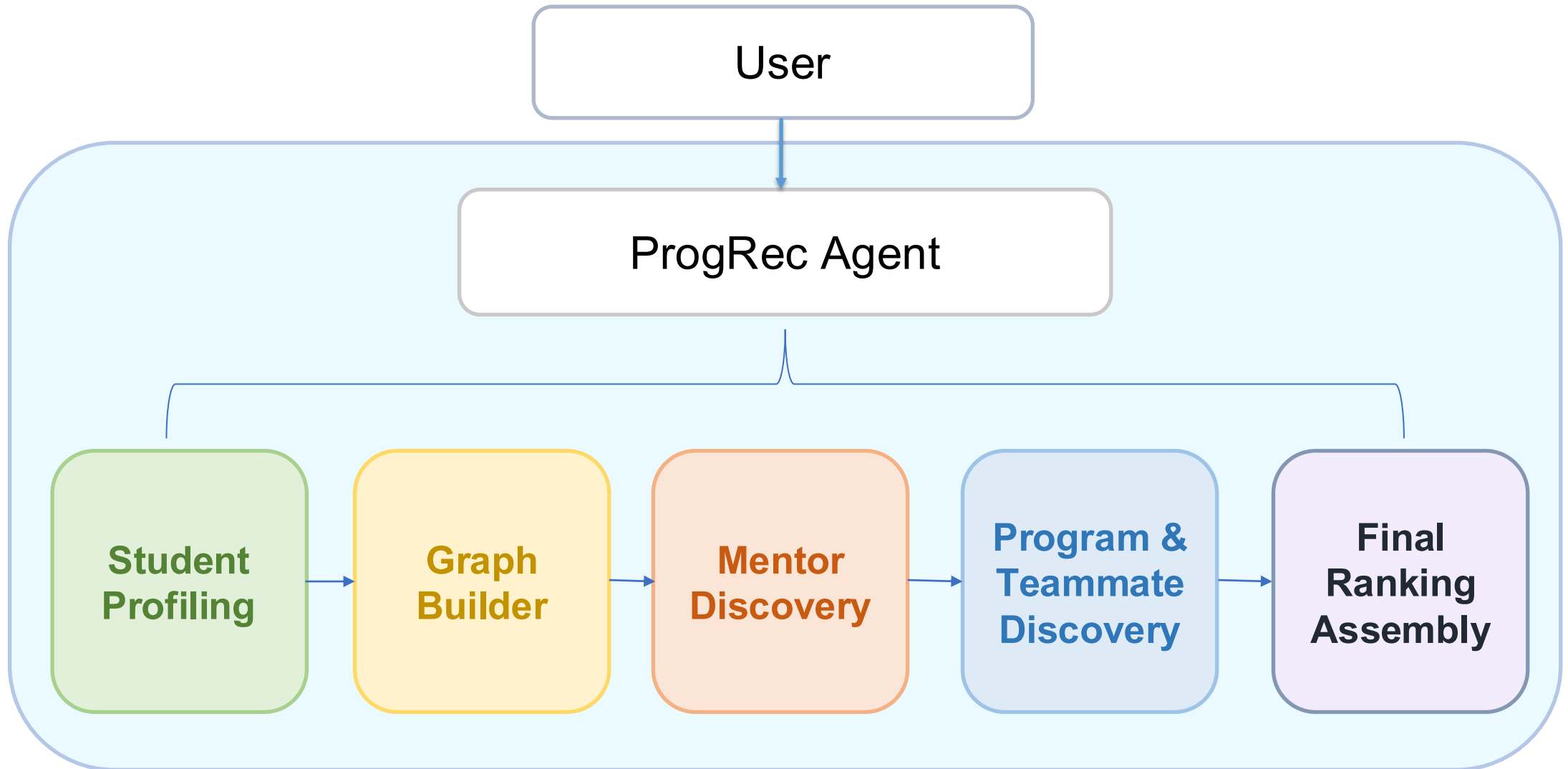
- Data Silos: Disconnected information
- Questions:
 - Who should guide my research journey?
 - Which project should I join?
 - Who can I collaborate with?
- Cold Start Trap: lack of academic network



Navigable research entry path



ProgRec Agent: An SNA-Driven Ecosystem



Skill 1: Student Profiling Skill

Name: Thomas Ibarra
Major: Biochemistry
Year: Senior
GPA: 2.83
Hobbies: Table Tennis, Photography
Unique Quality: Wildlife conservationist
Story: [2,000–3,300 chars of narrative text]
"Thomas had always been fascinated by wildlife. He spent summers documenting endangered species..."

```
"student_id": "thomas-ibarra-00000",  
"grade": "Senior",  
"major": "Biochemistry",  
"skills": ["laboratory techniques",  
           "molecular analysis",  
           "data analysis",  
           "scientific writing"],  
"interests": ["photography", "visual arts",  
              "wildlife conservation", "ecology"],  
"experience_summary": "Wildlife conservationist  
advocating for protection of endangered species.",  
"availability": "low"
```



23,236 Synthetic Undergraduate Profiles



Fields (11)

Name, Age, Sex, Major, Year, GPA,
Hobbies (2 items), Country,
State/Province, Unique Quality, Story

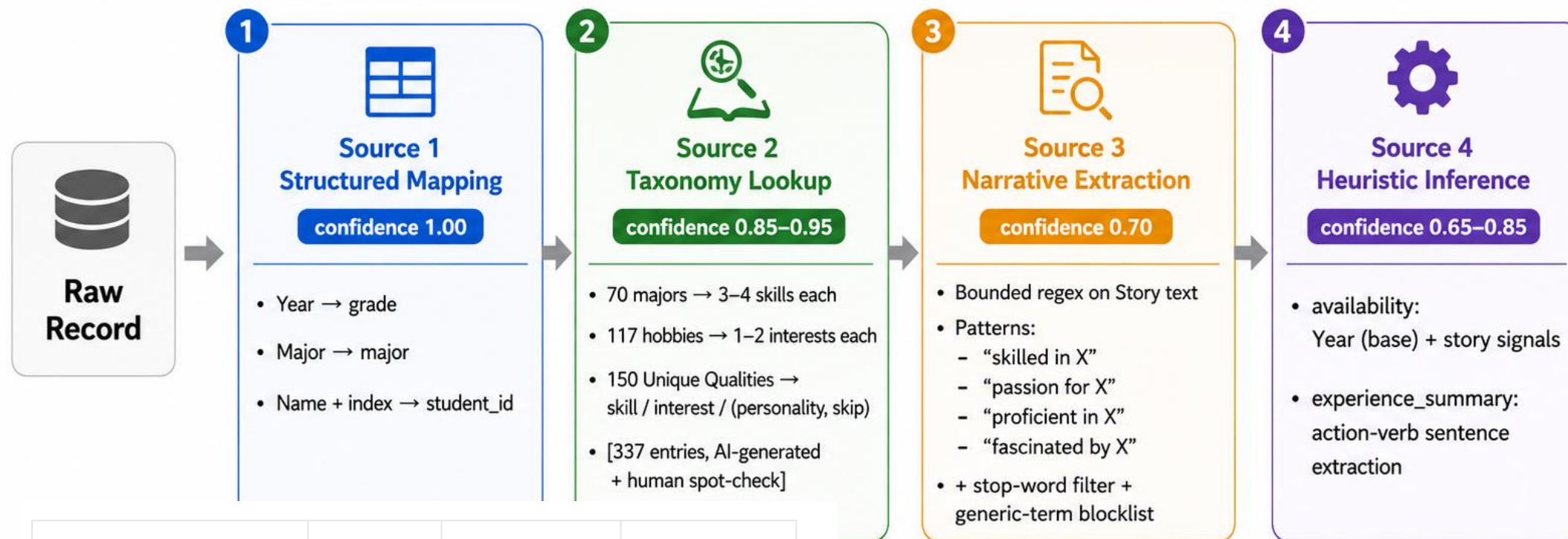


Key Challenge

No direct fields for research skills,
interests, project experience,
or availability.

- > No explicit skills, no research interests,
- > No availability — everything is buried in text.

Skill 1: Student Profiling Skill

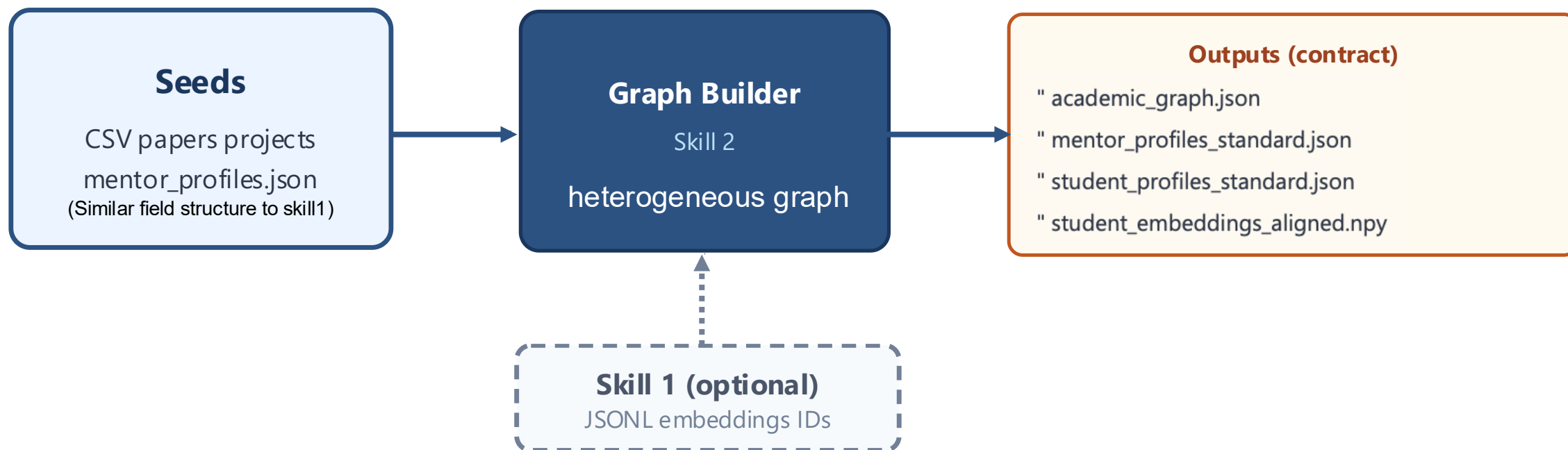


Configuration	Avg Skills	Avg Interests	Unique Terms
Baseline (raw fields only)	1.0	2.0	70
+ Taxonomy mapping	4.0	4.0	183
+ UQ mapping	5.21	4.54	322
Full pipeline	5.54	5.73	446

Skill 2: Academic Graph Builder Skill Pipeline

Inputs: seeded profiles + papers/projects (+ optional Skill 1 layer)

Outputs: contract files for later skills—no need to import Skill 2 code



Heterogeneous Graph: Nodes, Edges and Construction

Nodes (5 types)

Mentor
Paper
Topic
Project
Student

One JSON bucket per type in
academic_graph.json



Edges (grouped)

Academic

authored, paper_topic,
mentor_topic, topic_similarity,
collaboration

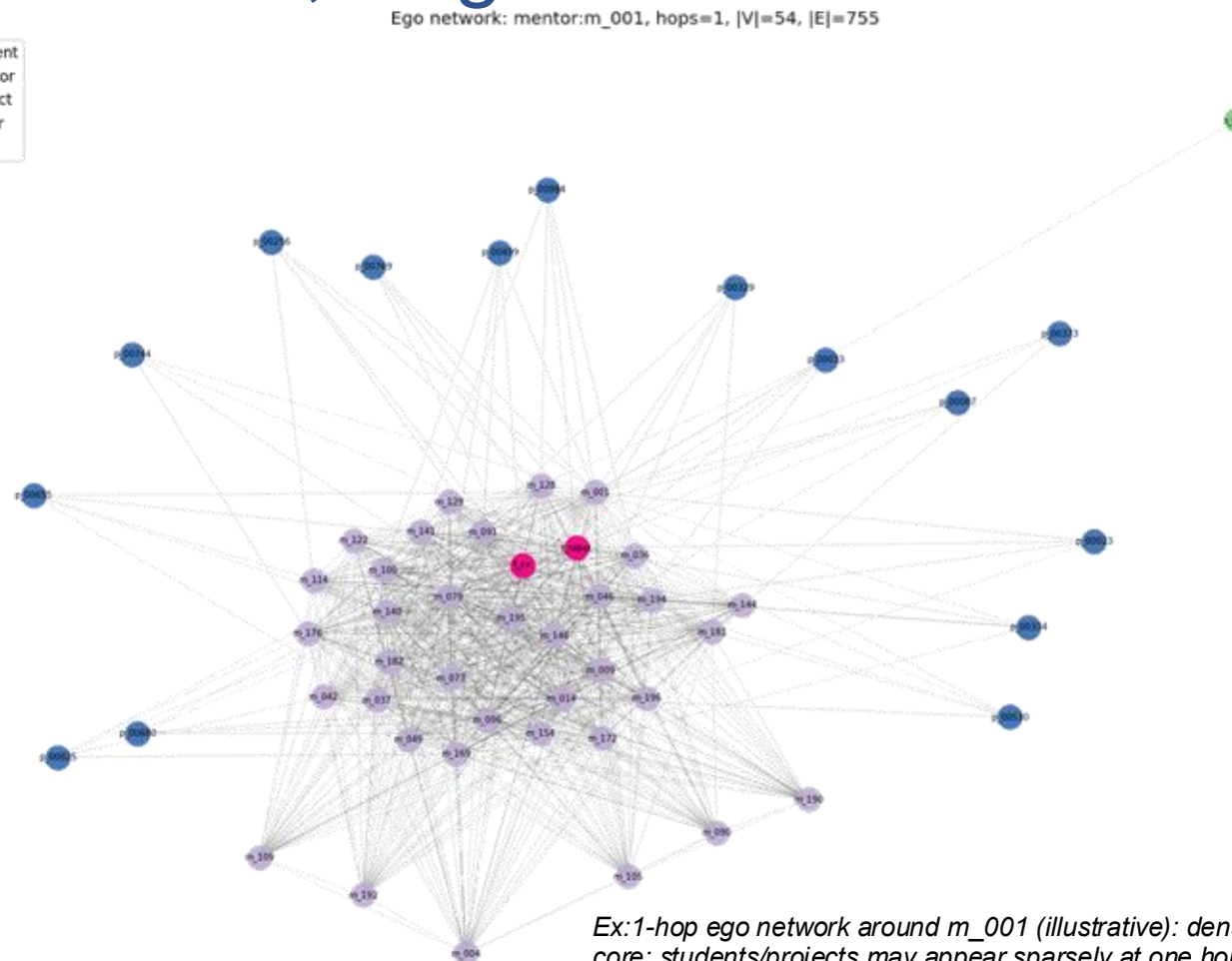
Campus

project_leads,
project_participation, advising

Peer

skill_complementarity,
shared_interest

Each edge carries type + weight
for downstream SNA / explanations



Ex: 1-hop ego network around m_001 (illustrative): dense mentor–paper–topic core; students/projects may appear sparsely at one hop

Mentor–topic links are aggregated via paper chains; student–student edges are capped at scale.
In alignment with Skill 1 mode, advising/participation are resynthesized for consistent student IDs.

Skill 2: Deliverables and Skill 1 Alignment

Deliverable	Purpose
<code>academic_graph.json</code>	One graph for analysis & explanations
<code>mentor_profiles_standard.json</code>	Mentor profiles ready for retrieval / embeddings
<code>student_profiles_standard.json(optional)</code>	Student profiles aligned with Skill 1
Student embeddings + ids(optional, both aligned)	Vectors line up with the exported student list

Skill 3 currently reads the Skill 1 student profiles/embeddings;
Skill 2's student bundle is an optional contract export for subset/order alignment with the graph and future single-folder handoff

Skill 3: Mentor Discovery Skill

Function:

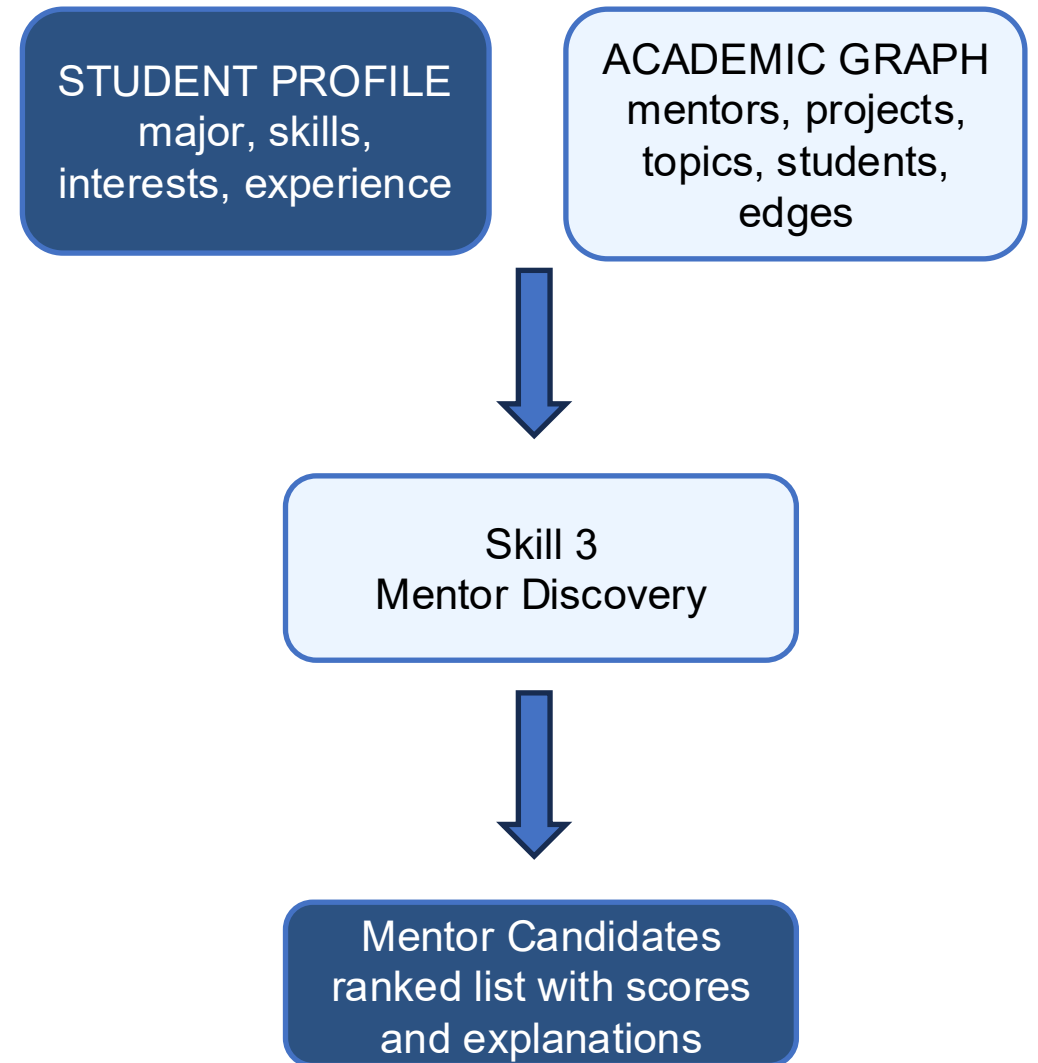
Retrieve Top-K mentor candidates for a student based on topic relevance and graph-aware signals.

Input:

- Student profile from Skill 1
- Mentor profiles from Skill 2
- Academic graph from Skill 2

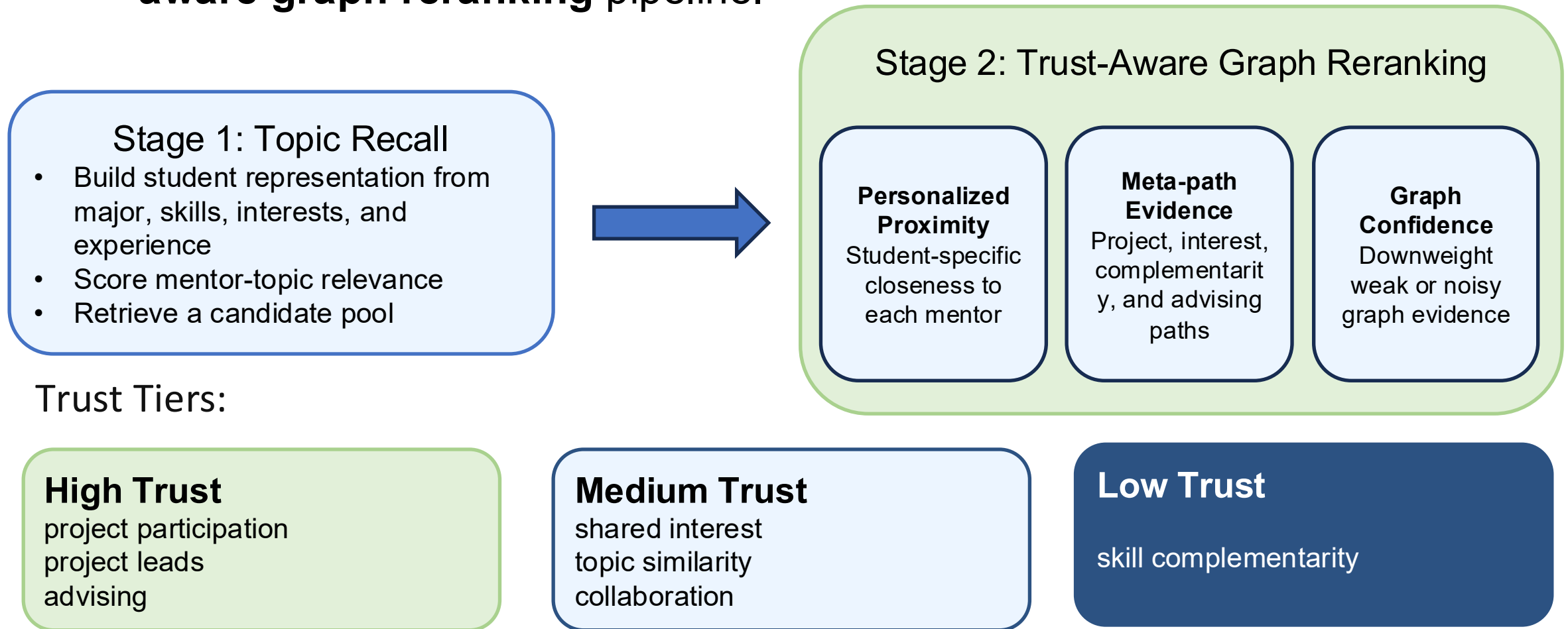
Output:

- Ranked mentor candidate list
- Topic score + graph score
- Explanation-ready fields for later skills



Skill 3: Core Upgrade

We upgrade a normal mentor retriever into a **topic recall + trust-aware graph reranking** pipeline.



Skill 3: Key Insight

Explainable

Path-level evidence and human-readable reasons

Trust-Aware

Weak graph signals are downweighted instead of treated equally

Scalable

Large graphs can switch to a lightweight subgraph fallback

Skill 4: Program & Teammate Discovery Skill

Function: Program and Teammate Recommendation

Input: Target student profile, Skill 2 academic graph, Skill 3 mentor candidates, Student candidate pool

Output: Related program, Recommended teammates



Skill 4: Program & Teammate Discovery Skill

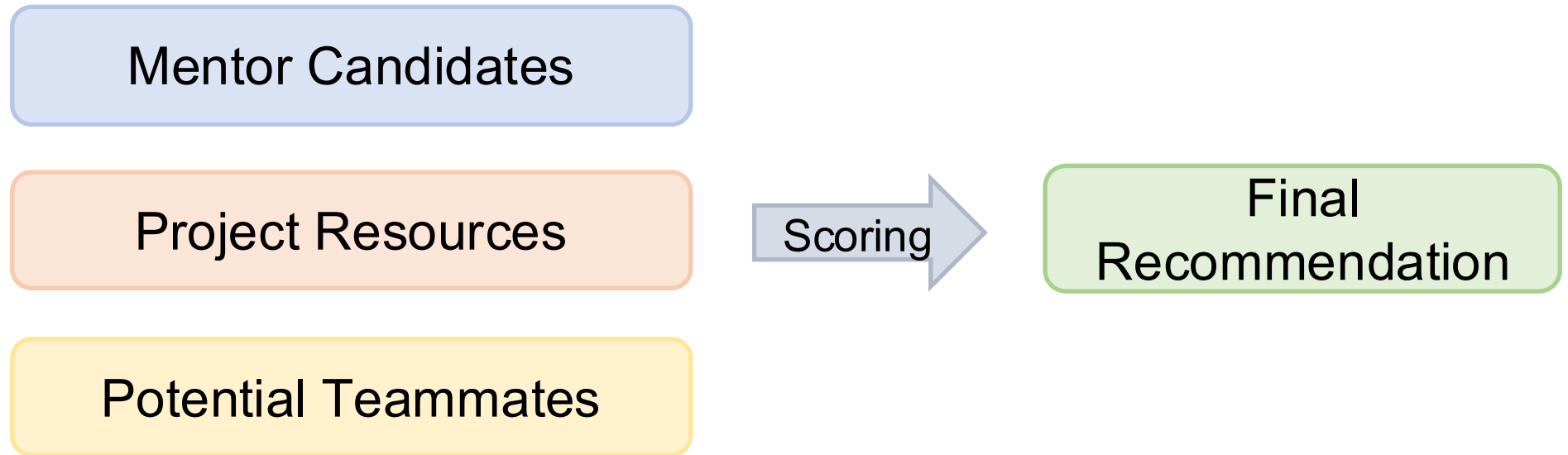
Function: Program and Teammate Recommendation

$$\begin{aligned}\text{Program Fit} = & 0.40 \times \text{Interest Match} \\ & + 0.30 \times \text{Skill Match} \\ & + 0.20 \times \text{Difficulty Match} \\ & + 0.10 \times \text{Mentor–Project Link}\end{aligned}$$

$$\begin{aligned}\text{Teammate Score} = & 0.35 \times \text{Shared Interest} \\ & + 0.35 \times \text{Skill Complementarity} \\ & + 0.10 \times \text{Availability} \\ & + 0.20 \times \text{Graph Relation}\end{aligned}$$

Skill 5: Social Ranking Skill

Function:



Key Method:

- Maximal Marginal Relevance (MMR) diversity re-ranking

$$MMR = \text{Arg} \max_{d_i \in R \setminus S} [\lambda \cdot \text{Sim}_1(d_i, Q) - (1 - \lambda) \cdot \max_{d_j \in S} \text{Sim}_2(d_i, d_j)]$$

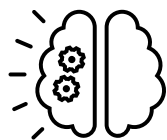
Skill 5: Social Ranking Skill

Highlight:

- Explainable Output: Reason
- Robustness: Cold Start: + cold start bonus

Mentor	Publication: empty	Topic < 2	
Student	Skills < 2	Experience_summary < 20 words	Interests < 2

Summary: From Data to Discovery



- Establish Research Persona
- Heterogeneous Graph
- SNA-based retrieval: topic recall + trust aware
- Executable and Explnable
- Re-ranking to Assemble the Final Recommendation